

Anderson localisation with self-avoiding-walks

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Zusammenfassung

Wir führen die Benutzung einiger $\text{\LaTeX}$ -Befehle und -Pakete anhand von Beispielen vor. Die Verwendung der hier empfohlenen Lösungen ist, mit Ausnahme des Titelblatts, nicht verpflichtend, sollte Ihre Arbeit aber erheblich vereinfachen.

Dieses Dokument spiegelt den Stand von 2021 wider; wenn Sie es im Jahr 2030 oder später noch lesen, sollten Sie sich vermutlich nach Alternativen für hier vorgestellte Lösungen umschauchen.

—Pavel Zorin-Kranich

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analogue of a random walk in a random environment. The model considers a d -dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump, it indicates the absence of quantum transport. This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.

Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.

The two important methods to prove Anderson localization are the *multiscale analysis method* (MSA), developed by Froehlich and Spencer [1] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. We will deal with the second one, which is more elementary and gives stronger results on *dynamical localization* [Chapter 2.2].

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic

concepts such as independence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schroedinger operator. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix (the kinetic part) and V_ω is a diagonal matrix with random entries. This operator acts on the Hilbert space

$$l^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \rightarrow \mathbb{C} : \sum_{n \in \mathbb{Z}^d} |u(n)|^2 < \infty\}$$

with inner product $\langle u, v \rangle = \sum_n \overline{u(n)}v(n)$.

1.1.1 The operator

The Discrete Laplacian. The kinetic term $-\Delta$ is the discrete analogue of the usual negative Laplacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in l^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := - \sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k), \quad k \in \mathbb{Z}^d$$

where $|\cdot|$ denotes the l^2 norm. The discrete Laplacian is per definition bounded and consequently self-adjoint [for proofs see Appendix A]. Just as the continuum Laplacian the Fourier transform blah blah blah.

The spectrum is absolutely continuous and as similarity with the continuous laplacian, it has plane waves as generalized eigenfunctions.

The Potential Term.

Definition 1.1 (i.i.d. random variables). $(\omega_v)_{v \in V}$ are identically distributed if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_v \in A) =$

$\mu(A)$ for all $v \in V$ and Borel sets $A \subset \mathbb{R}$. $(\omega_v)_{v \in V}$ are independent if

$$\mathbb{P}(\omega_{v_1} \in A_1, \dots, \omega_{v_l} \in A_l) = \prod_{k=1}^l \mathbb{P}(\omega_{v_k} \in A_k) = \prod_{k=1}^l \mu(A_k)$$

for each finite subset $\{v_1, \dots, v_l\}$ of V and arbitrary Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

We assume that the distribution μ of ω_v is absolutely continuous with density ρ where ρ is bounded with compact support, i.e.,

$$\mu(B) = \int_B \rho(u) du \quad \text{for } B \subset \mathbb{R} \text{ Borel, } \rho \in L_0^\infty(\mathbb{R}).$$

Physically, V_ω represents the random electric potential created by nuclei at the sites $v \in \mathbb{Z}^d$ **ovviamente il testo da modificare in base al resto delle notazioni** .

$\lambda > 0$ is the disorder parameter

1.1.2 Self-adjointness

1.1.3 The spectrum of H_ω .

- in $\mathbb{R}$
- bounded $[0, 4d] + \text{supp}\mu$
- spectral division (any use of the spectral theorem should be sent to the Appendix)
- RAGE theorem (or maybe directly in the Appendix or chapter 2)

Dalla teoria spettrale per operatori autoaggiunti su uno spazio di Hilbert $\mathcal{H}$, lo spazio si decompone univocamente nella somma diretta di sottospazi invarianti ortogonali:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_{sc} \oplus \mathcal{H}_{ac} \tag{1.1}$$

a cui corrispondono rispettivamente lo spettro puramente puntuale $\sigma_{pp}(H)$, singolarmente continuo $\sigma_{sc}(H)$ e assolutamente continuo $\sigma_{ac}(H)$, con $\sigma(H) = \sigma_{pp}(H) \cup \sigma_{sc}(H) \cup \sigma_{ac}(H)$.

L'interpretazione fisica di questi sottospazi è chiarita dal fondamentale **Teorema RAGE** (Ruelle-Amrein-Georgescu-Enss):

Theorem 1.2 (RAGE). *Sia H un operatore di Schrödinger su $l^2(G)$ e $\mathcal{H}_c = \mathcal{H}_{ac} \oplus \mathcal{H}_{sc}$ il sottospazio spettrale continuo. Allora:*

1. $\phi \in \mathcal{H}_{pp}$ se e solo se per ogni $\epsilon > 0$ esiste un raggio $R < \infty$ tale che:

$$\sup_{t \in \mathbb{R}} \|\chi_{(|x| > R)} e^{-itH} \phi\| < \epsilon \iff \lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{(|x| > R)} e^{-itH} \phi\| = 0$$

ovvero lo stato ϕ rimane confinato in regioni compatte uniformemente nel tempo (stato legato).

2. $\psi \in \mathcal{H}_c$ se e solo se per ogni $R > 0$:

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{(|x| \leq R)} e^{-itH} \psi\| dt = 0$$

ovvero la probabilità che lo stato permanga in qualsiasi regione limitata decade in media temporale a zero (stati di scattering / trasporto).

1.2 Main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:

The aim of this thesis is to prove localisation on the whole spectrum of H_ω
...

Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.

potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei risultati con Frohlich e spencer e con Aizenman e Molchanov.

One of the main results concerns the values of λ

- $\lambda \gg 1$ and $\lambda = 0$
- general case for $d = 1, d \geq 2$

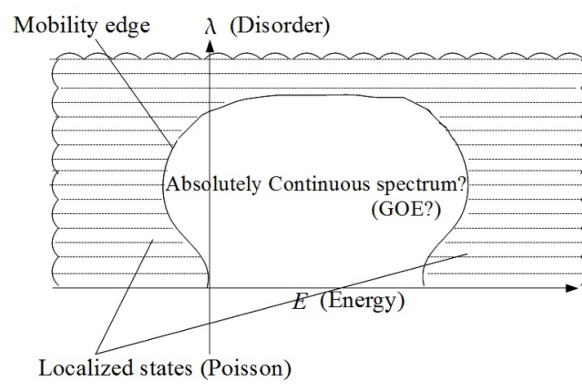


Figure 1: Aizenman figure

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 spectral localisation

Definition 2.1 (Spectral Localization). Sia $I \subset \mathbb{R}$ un intervallo aperto. Un operatore di Schrödinger casuale H_ω presenta **localizzazione spettrale** in I se, con probabilità 1 (quasi certamente), lo spettro continuo in I è vuoto e lo spettro è puramente puntuale:

$$\sigma_c(H_\omega) \cap I = \emptyset \quad \text{e} \quad \sigma(H_\omega) \cap I \subset \sigma_{pp}(H_\omega) \quad \text{quasi certamente.}$$

Definition 2.2 (Exponential Spectral Localization).

$$|\phi_{\omega,n}(x)| \leq C_{\omega,n} e^{-\mu|x-x_{n,\omega}|} \quad (2.1)$$

Remark 2.3 (Spettro puntuale denso). closure set of eigenvalues

2.2 dynamical localisation

Definition 2.4.

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k \rangle| \right) \leq C e^{-\mu|j-k|}$$

for all φ with compact support.

further explanation of the meaning

2.2.1 finite moments

Proposition: *Dynamical localization implies that all moments of the position operator are bounded in time, i.e., for all $p > 0$ and all finitely supported $\psi \in \ell^2(\mathbb{Z}^d)$,*

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \psi \| < \infty \quad \text{almost surely,} \quad (2.2)$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$.

Proof. To see how (1) follows from dynamical localization, assume that

$\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned} \| |X|^p e^{-ith\omega} \psi \|^2 &= \sum_j |\langle e_j, |X|^p e^{-ith\omega} \psi \rangle|^2 \\ &= \sum_j |j|^{2p} \left| \sum_{|k| \leq R} \langle e_j, e^{-ith\omega} e_k \rangle \psi(k) \right|^2 \\ &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} |\langle e_j, e^{-ith\omega} e_k \rangle|^2 \|\psi\|^2, \end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-ith\omega} e_k \rangle|^2$ (as this number is bounded by 1) and then take expectations to get

$$\begin{aligned} \mathbb{E} \left(\sup_t \| |X|^p e^{-ith\omega} \psi \|^2 \right) &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} \mathbb{E} \left(\sup_t |\langle e_j, e^{-ith\omega} e_k \rangle| \right) \|\psi\|^2 \\ &\leq C \sum_j \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\ &< \infty. \end{aligned}$$

□

2.3 From dynamical to spectral localisation

Theorem 2.5 (Localizzazione Dinamica $\implies$ Localizzazione Spettrale). *Se H_ω esibisce localizzazione dinamica in un intervallo aperto I , allora H_ω presenta quasi certamente solo spettro puramente puntuale in I ($\sigma_c(H_\omega) \cap I = \emptyset$).*

Schema della dimostrazione via RAGE. Applicando il Teorema RAGE alla proiezione continua $P_{cont}(H)\psi$ per uno stato ψ a supporto compatto in $\{|y| \leq r\}$:

$$\mathbb{E}(\|P_{cont}(H)\psi\|^2) \leq \lim_{R \rightarrow \infty} \frac{1}{T} \int_0^T dt \sum_{|x| \geq R, |y| \leq r} \mathbb{E}(|\langle \delta_x, e^{-itH} \delta_y \rangle|) \|\psi\|^2$$

Per localizzazione dinamica, la somma $\sum_{|x| \geq R} e^{-\mu|x-y|}$ tende a zero quando $R \rightarrow \infty$. Dunque $P_{cont}(H)\psi = 0$ quasi certamente su un insieme denso di stati ψ , da cui $\sigma_c(H_\omega) \cap I = \emptyset$. □

2.4 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t) = \left\| |x| e^{-itH_\omega} P_{H_\omega \in [a,b]} \phi_0 \right\|^2 \quad (5)$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av} , then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ will be proportional to t^2 for large t .

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ is bounded uniformly in t . It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t \rightarrow \infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0 , as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t \rightarrow \pm\infty} \frac{\langle x^2 \rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \quad \text{for all } \delta > 0, \quad (6)$$

for a large class of compactly supported initial conditions ϕ_0 . *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n , le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

3 Fractional moments

3.1 The green function

The *resolvent operator* is defined as:

$$G(z) = (H - z)^{-1}$$

3.2 First localisation criteria

Theorem 3.1 (Aizenman–Molchanov, 1993). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}(|G_{\omega,\lambda}(x, y; z)|^s) \leq C e^{-\mu|x-y|} \quad (3.1)$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$.

Lemma 3.2 (A priori bound). *There exists a constant $C_1 = C_1(s, \rho) < \infty$ such that*

$$\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x, y; z)|^s) \leq C_1 \lambda^{-s} \quad (3.2)$$

for all $x, y \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Here

$$\mathbb{E}_{x,y}(\dots) = \int \int \dots \rho(\omega_x) d\omega_x \rho(\omega_y) d\omega_y$$

is the conditional expectation with $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x,y\}}$ fixed. After averaging over ω_x and ω_y the bound in (1) does not depend on the remaining random parameters. Thus we get also that

$$\mathbb{E}(|G_{\omega,\lambda}(x, y; z)|^s) \leq C_1 \lambda^{-s}.$$

Proof. We first prove (1) for the case $x = y$, which demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x , we can separate the ω_x and $\hat{\omega}$ dependence of $H_{\omega,\lambda}$ as

$$H_{\omega,\lambda} = H_{\hat{\omega},\lambda} + \lambda \omega_x P_{e_x}.$$

The resolvent identity yields

$$(H_{\omega,\lambda} - z)^{-1} = (H_{\hat{\omega},\lambda} - z)^{-1} - \lambda \omega_x H_{\hat{\omega},\lambda} (H_{\omega,\lambda} - z)^{-1} P_{e_x} (H_{\omega,\lambda} - z)^{-1}. \quad (3.3)$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_{\omega,\lambda}(x, x; z) = G_{\hat{\omega},\lambda}(x, x; z) - \lambda \omega_x G_{\hat{\omega},\lambda}(x, x; z) G_{\omega,\lambda}(x, x; z) \quad (3.4)$$

or

$$G_{\omega,\lambda}(x, x; z) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{G_{\hat{\omega},\lambda}(x, x; z)}. \quad (3.5)$$

Note that the latter is well-defined since one can easily check the Herglotz property $\text{Im } G_{\hat{\omega},\lambda}(x, x; z) / \text{Im } z > 0$ of the Green function.

The important fact is that a is a complex number which does not depend on ω_x . Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$, we find that

$$\mathbb{E}_x(|G_{\omega,\lambda}(x, x; z)|^s) \leq \frac{\|\rho\|_\infty}{\lambda^s} \int_{\text{supp } \rho} \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \leq \frac{C(\rho, s)}{\lambda^s}, \quad (3.6)$$

with $C(\rho, s)$ independent of λ and a , and thus independent of $\hat{\omega}$, z and x .

The proof of (1) for $x \neq y$ is based on the same idea, replacing the rank-one-perturbation arguments above with rank-two-perturbation arguments. Write $\omega = (\hat{\omega}, \omega_x, \omega_y)$, $P = P_{e_x} + P_{e_y}$ and

$$H_{\omega,\lambda} = H_{\hat{\omega},\lambda} + \lambda \omega_x P_{e_x} + \lambda \omega_y P_{e_y}.$$

Using the resolvent identity similar to above one arrives at

$$P(H_{\omega,\lambda} - z)^{-1}P = \left(A + \lambda \begin{pmatrix} \omega_x & 0 \\ 0 & \omega_y \end{pmatrix} \right)^{-1}, \quad (3.7)$$

where

$$A = (P(H_{\hat{\omega},\lambda} - z)^{-1}P)^{-1},$$

both to be read as identities for 2×2 -matrices in the range of P . This is a special case of the Krein formula which characterizes the resolvents of finite-rank perturbations of general self-adjoint operators. For the matrix A one can check that $\text{Im } A = \frac{1}{2i}(A - A^*) < 0$ if $\text{Im } z > 0$ and $\text{Im } A > 0$ if $\text{Im } z < 0$. It is also independent of ω_x and ω_y .

Using that $G_{\omega,\lambda}(x, y; z)$ is one of the matrix-elements of $P(H_{\omega,\lambda} - z)^{-1}P$,

we find

$$\begin{aligned}
\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x,y;z)|^s) &\leq \mathbb{E}_{x,y} \left(\left\| \left(A + \lambda \begin{pmatrix} \omega_x & 0 \\ 0 & \omega_y \end{pmatrix} \right)^{-1} \right\|^s \right) \\
&= \lambda^{-s} \mathbb{E}_{x,y} \left(\left\| \left(-\frac{1}{\lambda} A - \begin{pmatrix} \omega_x & 0 \\ 0 & \omega_y \end{pmatrix} \right)^{-1} \right\|^s \right) \\
&\leq \frac{\|\rho\|_\infty^2}{\lambda^s} \int_{-r}^r \int_{-r}^r \left\| \left(-\frac{1}{\lambda} A - \begin{pmatrix} \omega_x & 0 \\ 0 & \omega_y \end{pmatrix} \right)^{-1} \right\|^s d\omega_x d\omega_y,
\end{aligned}$$

where $[-r, r]$ is an interval containing $\text{supp } \rho$. In the double integral we change variables to

$$u = \frac{1}{2}(\omega_x + \omega_y), \quad v = \frac{1}{2}(\omega_x - \omega_y),$$

which gives a Jacobian factor of 2. As $(\omega_x, \omega_y) \in [-r, r]^2$ implies $(u, v) \in [-r, r]^2$ we arrive at the bound

$$\begin{aligned}
\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x,y;z)|^s) &\leq \frac{2\|\rho\|_\infty^2}{\lambda^s} \int_{-r}^r \int_{-r}^r \left\| \left(-\frac{1}{\lambda} A + \begin{pmatrix} -v & 0 \\ 0 & v \end{pmatrix} - uI \right)^{-1} \right\|^s du dv \\
&\leq \frac{4r\|\rho\|_\infty^2}{\lambda^s} C(r, s) = \frac{C(s, \rho)}{\lambda^s}.
\end{aligned}$$

That the latter bound is uniform in x, y and z as well as in the random parameters $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x, y\}}$ follows from the fact that the matrix

$$-\frac{1}{\lambda} A + \begin{pmatrix} -v & 0 \\ 0 & v \end{pmatrix}$$

has either positive or negative imaginary part and the following general result:

For every $s \in (0, 1)$ and $r > 0$ there exists $C(r, s) < \infty$ such that

$$\int_{-r}^r \|(B - uI)^{-1}\|^s du \leq C(r, s) \quad (3.8)$$

for all 2×2 -matrices B such that either $\text{Im } B \geq 0$ or $\text{Im } B \leq 0$.

Starting with the observation that, by Schur's Theorem, B may be assumed upper triangular, we also may assume without loss that $\text{Im } B \geq 0$.

Thus

$$B = \begin{pmatrix} b_{11} & b_{12} \\ 0 & b_{22} \end{pmatrix} \quad (3.9)$$

and

$$(B - uI)^{-1} = \begin{pmatrix} \frac{1}{b_{11}-u} & -\frac{b_{12}}{(b_{11}-u)(b_{22}-u)} \\ 0 & \frac{1}{b_{22}-u} \end{pmatrix}. \quad (3.10)$$

The bound (7) follows if we can establish a corresponding fractional integral bound for the absolute value of each entry of (9) separately. For the diagonal entries this is obvious.

We bound the upper right entry of (9) by

$$\left| \frac{b_{12}}{(b_{11}-u)(b_{22}-u)} \right| \leq \frac{|b_{12}|}{|\operatorname{Im}((b_{11}-u)(b_{22}-u))|} = \frac{1}{\left| u \frac{\operatorname{Im} b_{11} + \operatorname{Im} b_{22}}{|b_{12}|} - \frac{\operatorname{Im}(b_{11}b_{22})}{|b_{12}|} \right|}. \quad (3.11)$$

The positive matrix

$$\operatorname{Im} B = \begin{pmatrix} \operatorname{Im} b_{11} & \frac{1}{2i} b_{12} \\ -\frac{1}{2i} \bar{b}_{12} & \operatorname{Im} b_{22} \end{pmatrix}$$

has positive determinant, i.e. $\det \operatorname{Im} B = \operatorname{Im} b_{11} \operatorname{Im} b_{22} - |b_{12}|^2/4 \geq 0$. We thus get

$$\left| \frac{\operatorname{Im} b_{11} + \operatorname{Im} b_{22}}{b_{12}} \right|^2 \geq \frac{2 \operatorname{Im} b_{11} \operatorname{Im} b_{22}}{|b_{12}|^2} \geq \frac{1}{2}.$$

The latter allows to conclude the required integral bound for (10). $\square$

3.2.1 Proof without SAW

4 SAW representation

4.1 SAW Definitions

4.2 SAW representation of G

- Theorem of the representation
 1. first prove that $G(x, y) = G(x, x) \sum G(x_1, y)$
 2. then the representation with the R_N
- put figure of the grid and the different paths

4.3 Proof with SAW

5 From fractional moments to dynamical localisation

Theorem 5.1. *Theorem final*

Proposition 5.2.

$$|\Im z| \mathbb{E} (|G(x, y; z)|^2) \leq C_1 \mathbb{E} (|G(x, y; z)|^s)$$

Proof of theorem 5.1.

□